

Zecpath AI Recruitment Platform

Technical Architecture & System Documentation

Document Version	1.0.0-PROD
Scope	Production Backend Topology, Subsystems, Data Models, & Operational Workflows
Classification	Internal Technical Engineering Documentation
Prepared by	RAJKUMAR PR

1. System Breakdown

1.1 Authentication & RBAC System

- **Identity Model:** Custom user model (CustomUser) extending Django's AbstractBaseUser, enforcing email-based identity with assigned roles (candidate, employer, admin).
- Stateless JWT Lifecycle:
 - **Access Token:** HMAC-SHA256 signed payload containing user_id, role, and claims with a short 15-minute time-to-live (TTL). Carried via Authorization: Bearer <access_token>.
 - **Refresh Token Strategy:** 7-day lifespan with Sliding Refresh Token Rotation (every refresh invalidates the prior token and generates a new pair).
- **Token Blacklisting:** Revoked and rotated refresh tokens are blacklisted in Redis to block token replay attacks.
- **Role-Based Access Control (RBAC):** Custom DRF permission classes (IsCandidateUser, IsEmployerUser, IsAdminUser) check user claims prior to view execution. Subscription-gated endpoints check active entitlements through a feature-access authorization layer.

1.2 Job Portal Core Requisition Lifecycle

- **Employer Management:** Authenticated employers create, update, or archive job postings. Jobs specify required competencies, minimum experience, location types, and salary brackets.
- **Public & Authenticated Search:** Multi-parameter search supporting keyword lookups, skill sets, and pagination.
- Application Funnel State Machine:

applied → screening → interview_scheduled → interview_completed → selected / rejected

State transitions are guarded; unauthorized jumps (e.g., transitioning directly from *applied* to *selected* without screening) are rejected with HTTP 400 Bad Request.

1.3 Asynchronous ATS (Applicant Tracking System) Engine

- **Cloud File Ingestion:** Large binary documents (PDF/DOCX resumes) are uploaded directly to an AWS S3 private bucket via short-lived pre-signed POST URLs, keeping high-bandwidth multi-part file streams off the web application server.
- **Asynchronous Text Extraction:** Upload completion triggers an asynchronous Celery task through a Redis message broker. Workers leverage stream-based PDF parsers (pdfplumber / Apache Tika) inside context-managed threads to prevent memory leaks.
- Vector Semantic Matching:
 - Extracts candidate competencies, titles, and quantified experience.
 - Transforms resume content and job requisitions into semantic dense vector embeddings.
 - Computes Cosine Similarity:

$$\text{Similarity} = (A \cdot B) / (\|A\| \|B\|)$$

- Normalizes output to a 0-100 scale and persists the result to core_application.match_score, enabling instant filtering via composite indexes.

1.4 AI Voice Automation & Real-Time Interview Pipeline

- **Telephony & Media Streams:** Bidirectional audio streaming over WebSockets via Twilio Media Streams or WebRTC audio channels.
- **Speech-to-Text (STT):** 16 kHz raw PCM audio frames are streamed to an STT worker (OpenAI Whisper / Fast-Whisper), transcribing spoken candidate responses into normalized text tokens.
- **Dynamic Contextual Questioning:** LLM prompt orchestrator processes candidate CV profile tokens, job requirements, and prior conversational context to dynamically synthesize the next targeted technical question.
- **Candidate Scoring Subsystem:** Transcribed responses are evaluated across three dimensions: Technical Accuracy (50%), Communication & Fluency (30%), and Contextual Relevance (20%).

- Composite Ranking Aggregator:

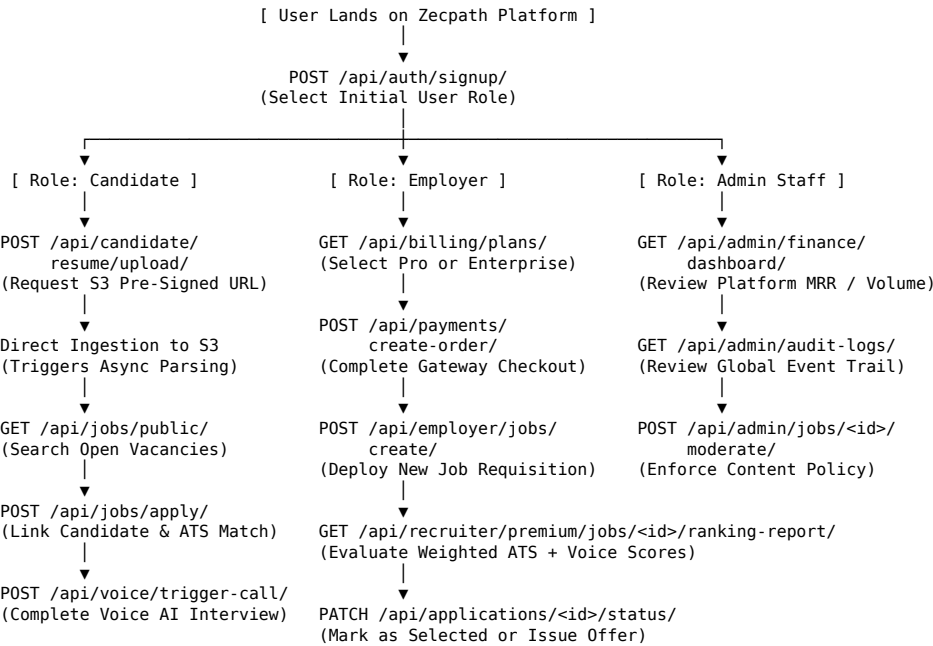
$$\text{Composite Score} = (0.50 \times \text{ATS Score}) + (0.50 \times \text{AI Interview Score})$$

1.5 SaaS Monetization & Production Deployment Architecture

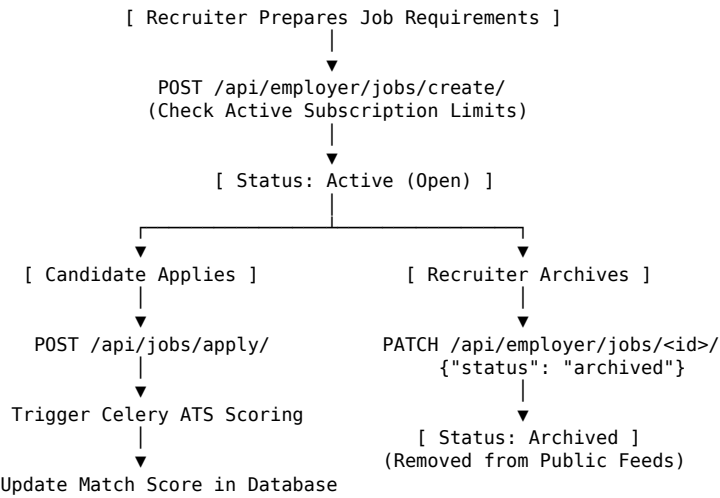
- **Monetization Engine:** Supports Free, Pro Recruiter (\$49/month), and Enterprise (\$199/month) plans.
- **Payment Gateways:** Stripe and Razorpay order generation with HMAC-SHA256 cryptographic signature validation on verification endpoints and asynchronous webhook deliveries.
- **Database Concurrency Controls:** Checkout and subscription state mutations execute inside `transaction.atomic()` blocks with row-level locks (`select_for_update()`) to prevent race conditions during concurrent billing events.
- Production Deployment Topology:
 - **Reverse Proxy:** Nginx 1.24 terminates TLS/SSL, handles static assets directly, proxies private media redirects, and forwards API traffic over Unix domain sockets.
 - **WSGI Daemon Cluster:** Gunicorn processes running DRF under systemd process supervision.
 - **Relational Storage:** PostgreSQL 16 tuned with connection pooling (`CONN_MAX_AGE=600`) and composite B-tree indexes.
 - **Cache & Broker:** Redis 7 cluster handling Celery message queues, API rate-limiting buckets, and 5-minute database query caches.

2. Flowcharts (System & Modules)

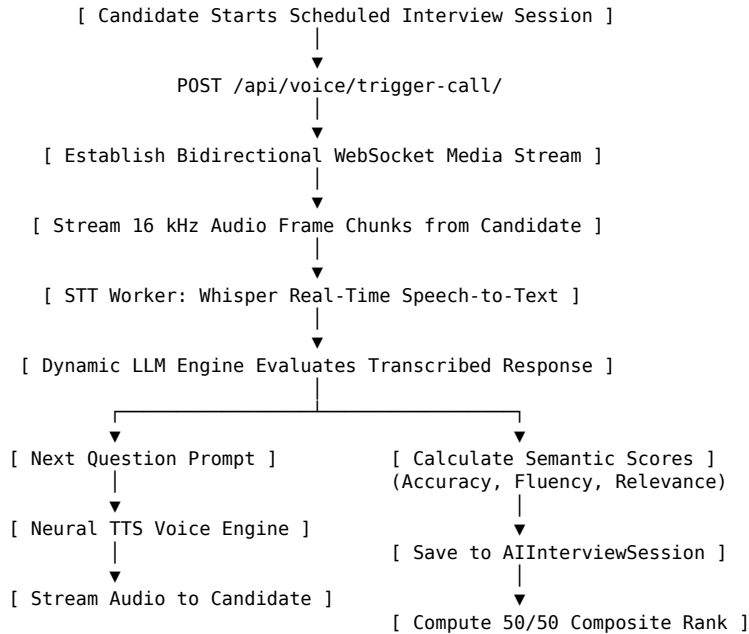
2.1 Complete End-to-End User Flow (Candidate, Employer, Admin)



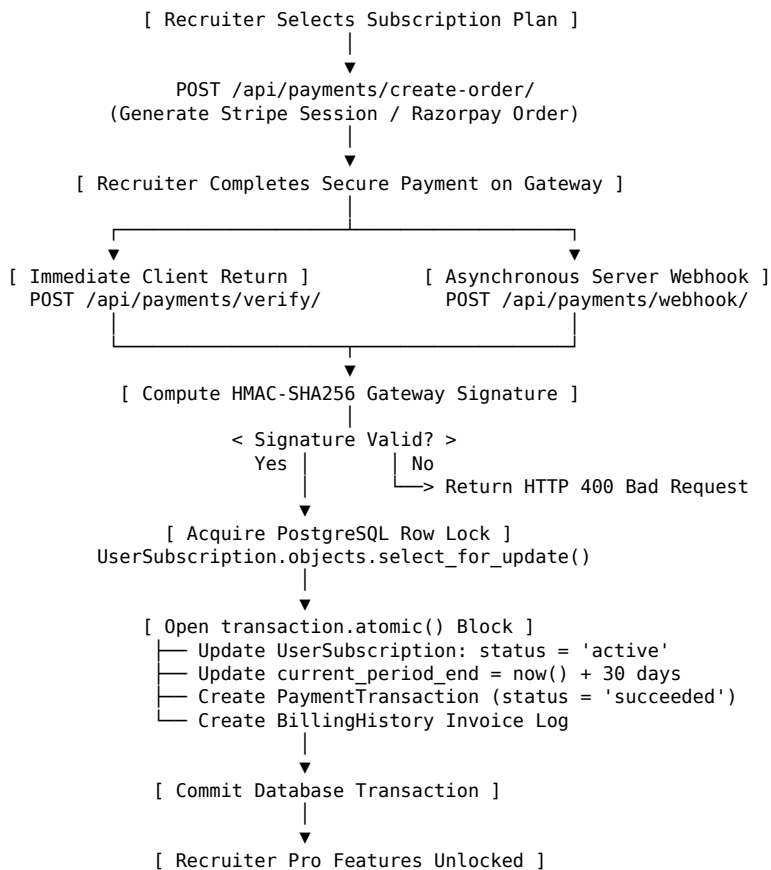
2.2 Job Requisition Lifecycle Flow



2.3 AI Voice Interview Operational Flow

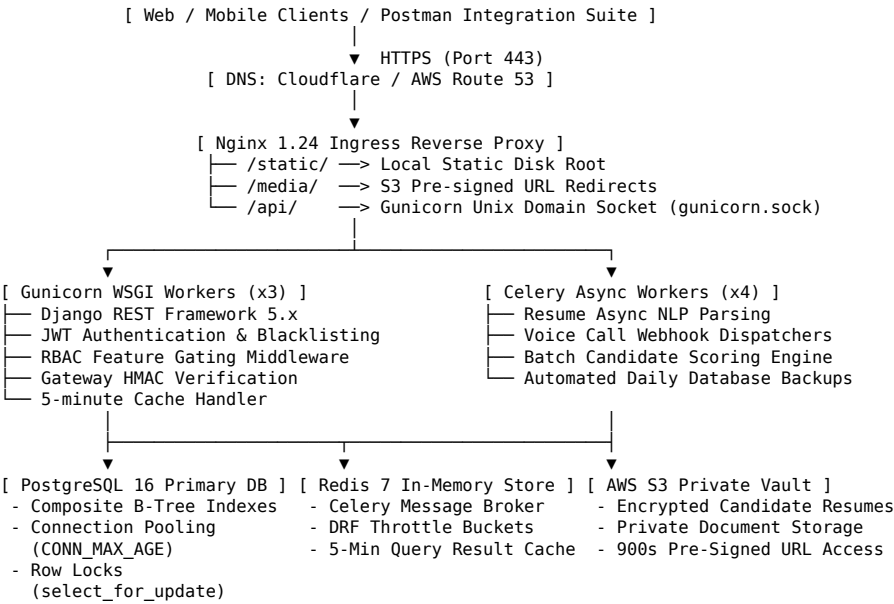


2.4 SaaS Monetization & Webhook Flow

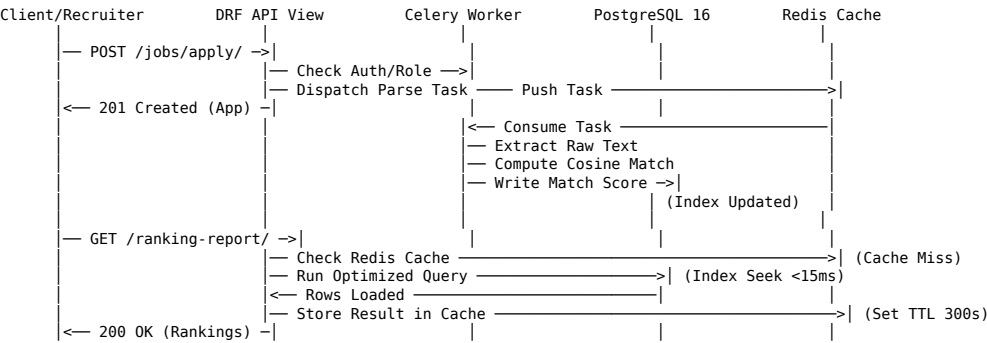


3. Architecture Diagrams

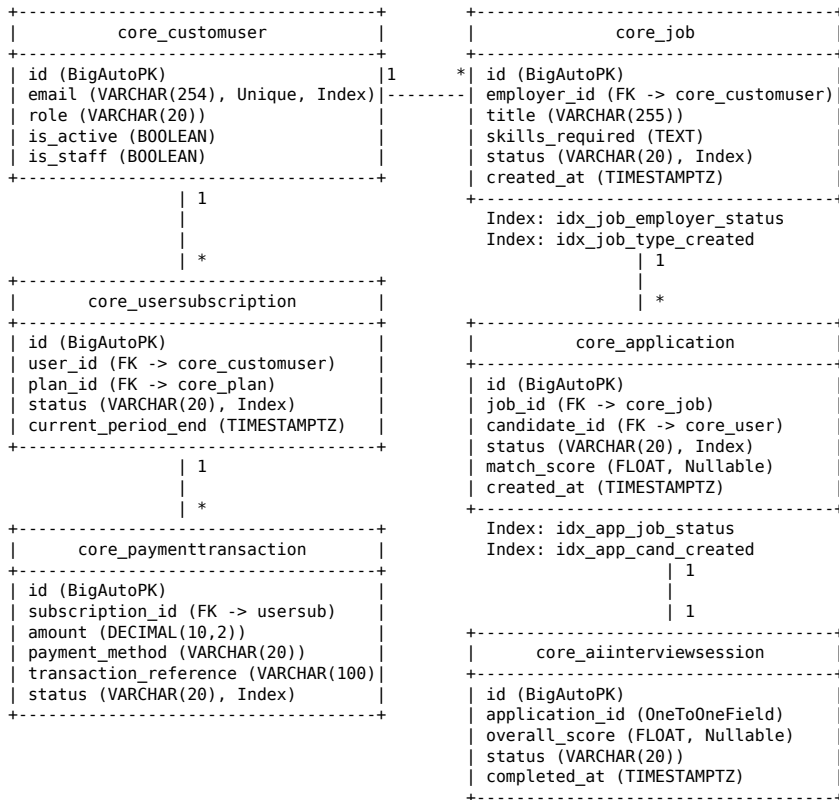
3.1 High-Level System Architecture Design



3.2 Service Interaction Diagram



3.3 Database Relational Schema & Indexing Topology



4. Documentation Writing: Module-Wise API Overview

Domain / Module	Method	Endpoint Route	Auth Scope	Description / Data Contract
Authentication	POST	/api/auth/signup/	AllowAny	Creates candidate or employer identity with salted Argon2/PBKDF2 passwords.
Authentication	POST	/api/auth/login/	AllowAny	Issues standard short-lived JWT access token (15m) and rotating refresh token (7d).
Authentication	POST	/api/auth/logout/	Authenticated	Revokes refresh token and registers its unique ID (JTI) into the Redis blacklist.
Cloud Storage	POST	/api/storage/resumes/presigned-upload/	Candidate	Generates temporary Amazon S3 pre-signed POST credentials for direct multipart client upload.
Cloud Storage	GET	/api/storage/resumes/secure-url/	Employer / Admin	Issues 900-second secure temporary link to download/view private candidate resumes.
Jobs & Portals	GET	/api/jobs/public/	AllowAny	Paginated public job feed with keyword search and category filtering.
Jobs & Portals	POST	/api/employer/jobs/create/	Employer (Gated)	Creates an active requisition; enforces active subscription plan limits.
ATS Engine	POST	/api/jobs/apply/	Candidate	Links candidate profile to target job, dispatches async text parsing and semantic scoring.
AI Interview	POST	/api/voice/trigger-call/	Candidate / Admin	Opens telephony/WebSocket media session for AI speech transcription and dynamic evaluation.
Recruiter Insights	GET	/api/recruiter/premium/jobs/<id>/ranking-report/	Employer (Pro)	Returns weighted composite rankings (ATS 50% + AI 50%) via optimized DB indexes.
SaaS Billing	POST	/api/payments/create-order/	Authenticated	Generates payment order via Stripe or Razorpay for Pro/Enterprise tier checkout.

Domain / Module	Method	Endpoint Route	Auth Scope	Description / Data Contract
SaaS Billing	POST	/api/payments/webhook/	AllowAny (HMAC)	Handles asynchronous payment success events with SHA256 HMAC verification.
Platform DevOps	GET	/api/load-test/ping/	AllowAny	Public synthetic routing probe returning health status and system timestamp.
Platform DevOps	GET	/api/qa/final-signoff/	Admin	Self-check verifying DB round-trip latency (<100ms) and subsystem readiness.
Architecture API	GET	/api/system/architecture-overview/	—	System architecture overview endpoint.

Prepared by **RAJKUMAR PR**